

BS – 101 Engineering Physics

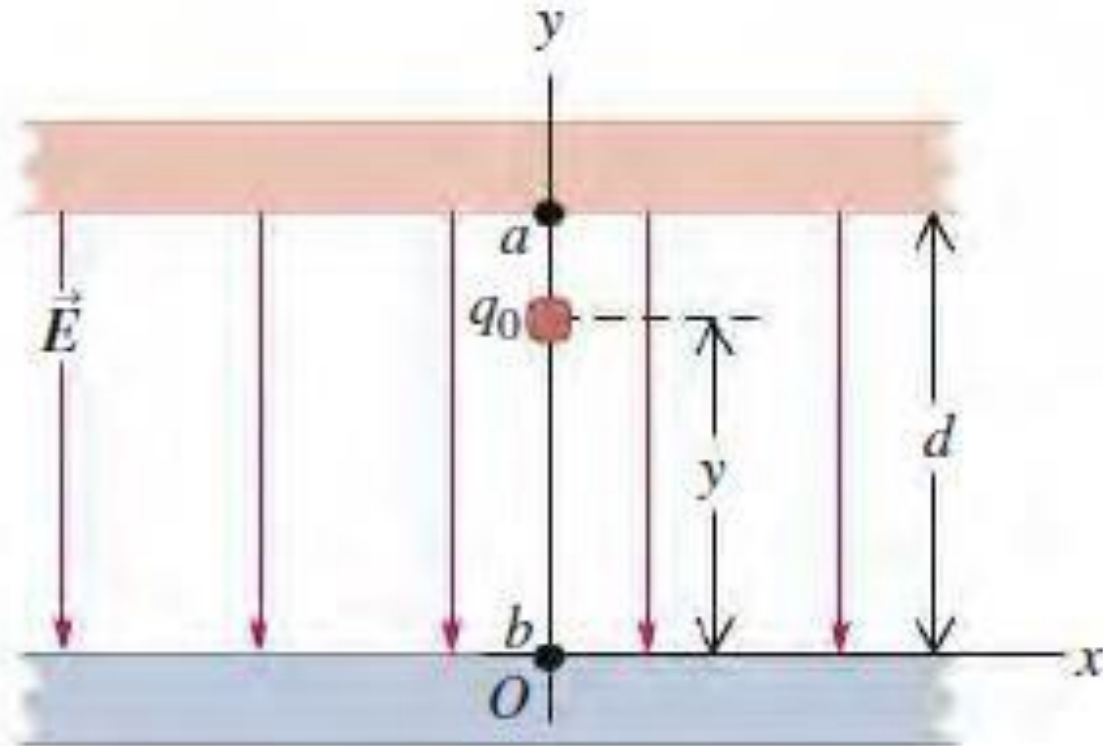
Lecture – 7

Engr. Safee Ullah



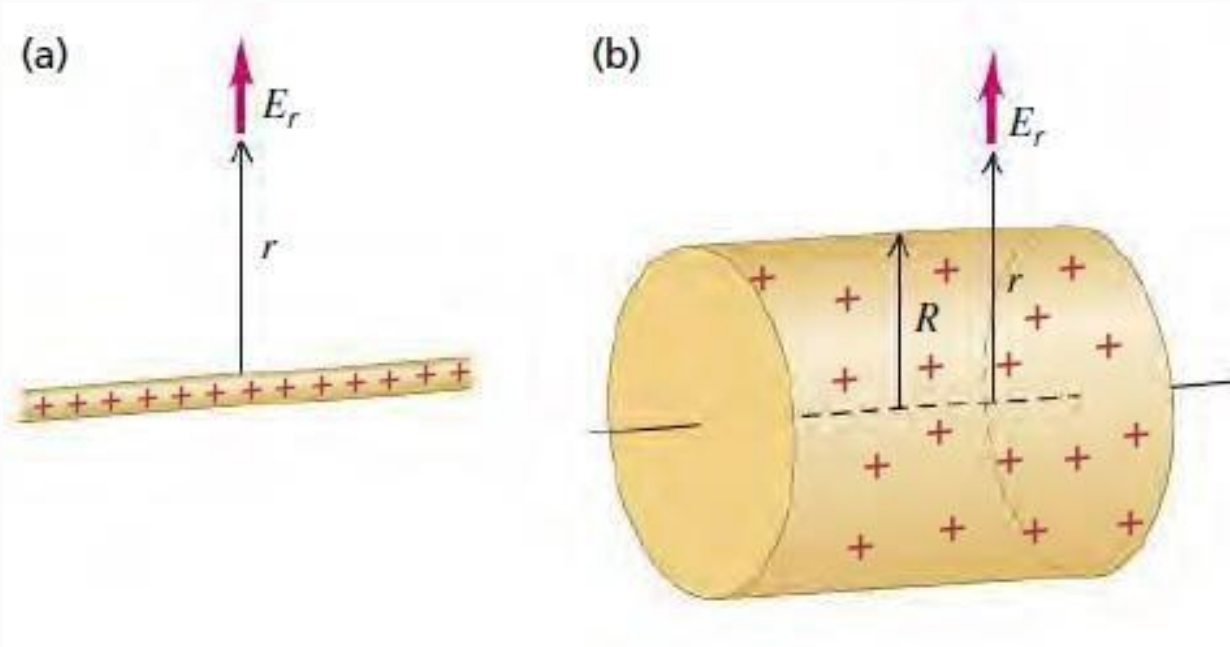
Oppositely Charged Parallel Plates

- Find the potential at any height y between the two oppositely charged parallel plates.



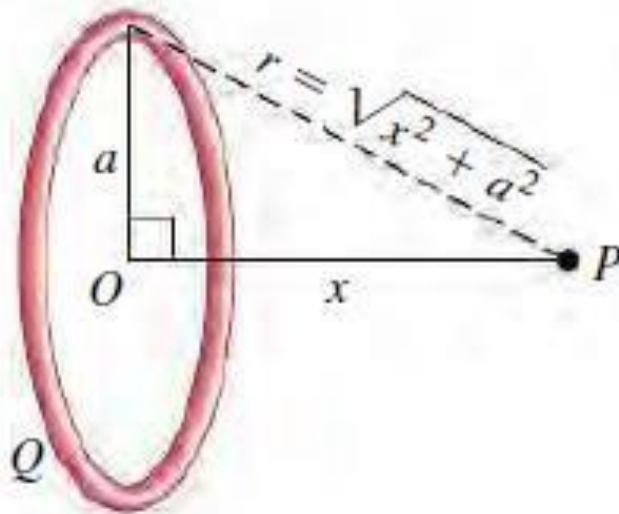
Charged Conducting Cylinder

- Find the potential at a distance r from a very long line of charge with linear charge density (charge per unit length) λ .



Charged Ring

- Electric charge Q is distributed uniformly around a thin ring of radius a . Find the potential at a point P on the ring axis at a distance x from the center of the ring.





Potential Gradient

- Derivations



Potential and Field of a Point Charge

- The potential at a radial distance r from a point charge q is $V = \frac{q}{4\pi\epsilon_0 r}$. Find the vector electric field from this expression for V .

Potential and Field of a Ring of Charge

- We found that for a ring of charge with radius a and total charge Q , the potential at a point P on the ring's symmetry axis a distance x from the center is:

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{x^2 + a^2}}$$

- Find the electric field at P .